

Dů. 3.2

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a)

Ověříme, že  $\text{psl}(1, x, x^2)$  je LN  $\rightarrow$  báze, podprostor

$\rightarrow$  uvidíme z tvrzení, že <sup>pokud</sup>  $\forall$  lineární kombinací prvků dostaneme

nulovou funkci ~~ž~~ jen v triviálním případě, pak je LN

$$a1 + bx + cx^2 = 0 \Leftrightarrow a, b, c = 0 \quad \checkmark$$

$\rightarrow$  Hledáme ON bázi:  $B = (u_1, u_2, u_3)$  podprostoru  $\text{span}\{1, x, x^2\}$

$\rightarrow$  pomocí GS ortogonalizace

$$u_1 = \frac{1}{\sqrt{\langle 1, 1 \rangle}} = \frac{1}{\sqrt{\int_1^5 1 \, dx}} = \frac{1}{2}$$

$$u_2 = \frac{w_2}{\sqrt{\langle w_2, w_2 \rangle}} \quad w_2 \text{ je } \text{projekce } x \text{ na } \text{span}\left\{\frac{1}{2}\right\}$$

$$w_2 = x - \left\langle \frac{1}{2}, x \right\rangle \frac{1}{2} = x - \frac{1}{2} \int_1^5 \frac{x}{2} \, dx = x - \frac{1}{2} \cdot 6 = \underline{x-3}$$

$$u_2 = \frac{(x-3)}{\sqrt{\int_1^5 (x-3)^2 \, dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \underline{\underline{\frac{\sqrt{3}}{4} (x-3)}}$$

$$u_3 = \frac{w_3}{\sqrt{\langle w_3, w_3 \rangle}}$$

$$w_3 = x^2 - \text{proj}_{\text{span} \left\{ \frac{1}{2}, \frac{\sqrt{3}}{4}(x-3) \right\}} x^2$$

$$w_3 = x^2 - \left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{2} - \left\langle \frac{(\sqrt{3}(x-3))}{4}, x^2 \right\rangle \frac{\sqrt{3}(x-3)}{4} =$$

$$= x^2 - \frac{1}{2} \int_1^5 \frac{x^2}{2} dx - \frac{\sqrt{3}(x-3)}{4} \int_1^5 \frac{\sqrt{3} x^2 (x-3)}{4} dx =$$

$$= x^2 - \frac{62}{6} - \frac{8\sqrt{3} \cdot \sqrt{3}(x-3)}{4} = x^2 - 6x + \frac{23}{3}$$

$$u_3 = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 \left(x^2 - 6x + \frac{23}{3}\right)^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{956}{45}}} = \underline{\underline{\frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right)}}$$

$$B = \left( \frac{1}{2}, \frac{\sqrt{3}}{4}(x-3), \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right) \right)$$


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b) OG projekce  $\sin x$  na  $\text{span} \{1, x, x^2\}$   
 vzhledem k tomu polynom  $ax^2 + bx + c$   $a, b, c \in \mathbb{R}$

→ Znale ON bázi  $B$  tedy

~~$$\text{projekce } \sin x = \left\langle \frac{1}{2}, \sin x \right\rangle \frac{1}{2} + \left\langle \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right), \sin x \right\rangle$$~~

$$\text{projekce } \sin x = \frac{1}{2} \left\langle \frac{1}{2}, \sin x \right\rangle + \frac{\sqrt{3}}{4} (x-3) \left\langle \frac{\sqrt{3}}{4} (x-3), \sin x \right\rangle +$$

$$+ \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right) \left\langle \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right), \sin x \right\rangle =$$

$$= \frac{1}{2} \cdot \int_1^5 \frac{\sin x}{2} dx + \frac{\sqrt{3}}{4} (x-3) \int_1^5 \frac{\sqrt{3} (x-3) \sin x}{4} dx +$$

$$+ \frac{3\sqrt{5}}{16} \int_1^5 \frac{3\sqrt{5} \left(x^2 - 6x + \frac{23}{3}\right)}{16} \cdot \sin(x) dx = \text{~~0,064~~}$$

~~$$0,064 + (x-3) \cdot (-0,647) + \left(x^2 - 6x + \frac{23}{3}\right) \cdot$$~~

$$\cdot (-0,053) = \underline{\underline{-0,053 x^2 - 0,329 x + 1,599}}$$

$$a = -0,053$$

$$b = -0,329$$

$$c = 1,599$$

c) aritme normu roddik  $\| \sin x - (ax^2 + bx + c) \|$

$$\| \sin x - (ax^2 + bx + c) \| = \sqrt{\int_1^5 (\sin x - (ax^2 + bx + c))^2 dx}$$

$$= \sqrt{\int_1^5 (\sin x + 0,053x^2 + 0,329x - 1,549)^2 dx} = 0,319$$